

Artificial Intelligence

Lesson 19: Bayes' Rule

Prior → Evidence → Update → Posterior

Belief update pipeline

Prior

$$P(F) = 0.05$$

Likelihood

$$P(A|F) = 0.90$$

Evidence

$$P(A) = 0.14$$

Posterior

$$P(F|A) = 0.321$$

Bayes' Rule updates belief by combining what was already plausible with how compatible the evidence is.



From conditioning to updating

L18 gave us conditional probability. L19 asks how evidence should change belief.

$$P(A | F)$$

Probability of an alert
if the component is faulty

Known condition → chance of evidence

$$P(F | A)$$

Probability the component is faulty
if we observe an alert

Observed evidence → updated belief

These are not the same question. Bayes' Rule helps us reverse the direction of conditioning.

Today's mission

Bayes' Rule gives a principled way to update belief after seeing evidence.

- 1 Apply Bayes' Rule to update a prior belief after observing evidence.
- 2 Distinguish prior, likelihood, evidence, and posterior probabilities.
- 3 Compute the evidence term by marginalization in simple two-hypothesis problems.
- 4 Explain how base rates and likelihoods jointly determine posterior belief.

Working rule: Compute → Interpret → Compare

Start with the prior

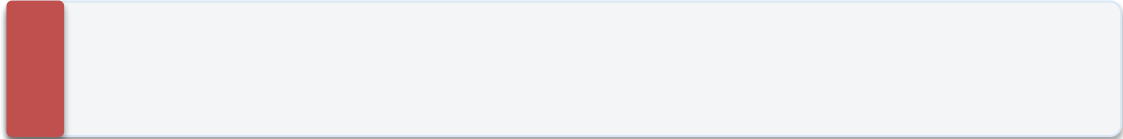
Before seeing new evidence, the agent begins with a prior belief about the world.

Inspection example

F = component is faulty

A = inspection system raises an alert

Prior belief before any alert


$$P(F) = 0.05 \quad \text{and} \quad P(\neg F) = 0.95$$

Only 5% of components are expected to be faulty
before the detector says anything.

Prior = what we believe before this new evidence arrives.

Evidence arrives

Now the inspection system produces new information.

Detector behavior

$$P(A | F) = 0.90$$

If a part is faulty, an alert appears 90% of the time.

False positives

$$P(A | \neg F) = 0.10$$

Even good parts trigger an alert 10% of the time.

Question

If the detector raises an alert, should we conclude the part is faulty with 90% probability?

**Likelihood tells us how compatible the evidence is with a hypothesis.
It does not yet answer the reverse question.**

Vocabulary for a Bayes update

Each quantity in Bayes' Rule plays a different role in reasoning.

Prior

$$P(F) = 0.05$$

What we believed before the alert.

Likelihood

$$P(A | F) = 0.90$$

How likely the evidence is if the hypothesis is true.

Alternative likelihood

$$P(A | \textit{not } F) = 0.10$$

How likely the same evidence is under another explanation.

Posterior

$$P(F | A) = ?$$

What we want after incorporating the alert.

Posterior belief should reflect both the prior and how selectively the evidence supports the hypothesis.

You Try: which probability is which?

Match each description to the correct probability expression.

- A. Probability the component is faulty before any alert
- B. Probability of an alert if the component is faulty
- C. Probability the component is faulty after an alert
- D. Total probability of observing an alert

1. $P(A)$

2. $P(F | A)$

3. $P(F)$

4. $P(A | F)$

**Discuss with a neighbor before revealing the answers:
which expressions represent prior, likelihood, evidence, and posterior?**

The same joint event can be written two ways

Bayes' Rule comes from equating two expressions for the same joint probability.

$$P(F, A) = P(A | F) P(F)$$

Start from the hypothesis, then ask about the evidence.

$$P(A, F) = P(F | A) P(A)$$

Start from the evidence, then ask about the hypothesis.

Because both expressions equal the same joint event,
they must equal one another.

$$P(A | F) P(F) = P(F | A) P(A)$$

Bayes' Rule

Solve the previous relationship for the posterior we care about.

$$P(F | A) = \frac{P(A | F) P(F)}{P(A)}$$

Posterior
 $P(F | A)$

Likelihood
 $P(A | F)$

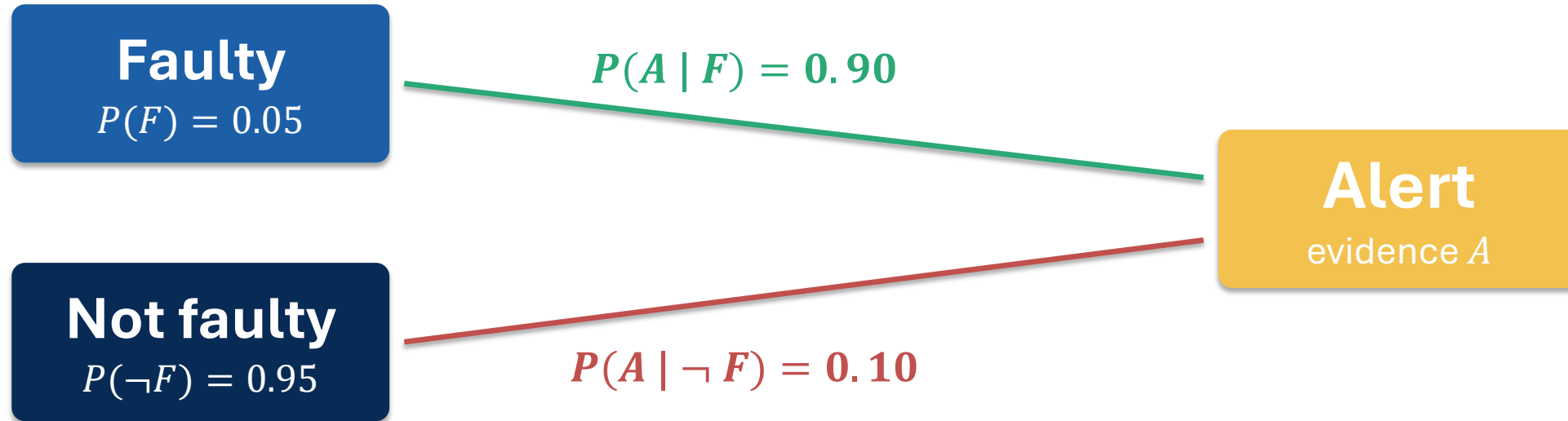
Prior
 $P(F)$

Evidence
 $P(A)$

A helpful memory aid: posterior $\propto$ likelihood $\times$ prior, then normalize using the evidence.

Where does the evidence term come from?

An alert can appear for more than one reason, so the denominator must account for all pathways.



Total evidence = sum over all ways the observed evidence could occur.

Marginalization computes the evidence term

Use L18's "sum out what you do not care about" idea to get $P(A)$.

$$P(A) = P(A, F) + P(A, \neg F)$$

Recall the product rule:

$$P(A, F) = P(F | A) P(A)$$

$$P(A) = P(A | F) P(F) + P(A | \neg F) P(\neg F)$$

First pathway

faulty and alert

$$P(A, F) = 0.90 \times 0.05$$

Second pathway

not faulty and alert

$$P(A, \neg F) = 0.10 \times 0.95$$

You Try: compute the evidence

Use the pathway expression to find the total probability of an alert.

$$P(A) = P(A, F) + P(A, \neg F)$$

$$P(A) = P(A | F) P(F) + P(A | \neg F) P(\neg F)$$

First pathway

faulty and alert

$$P(A, F) = 0.90 \times 0.05$$

Second pathway

not faulty and alert

$$P(A, \neg F) = 0.10 \times 0.95$$

Stop and solve:

What is the total probability that the detector raises an alert?

1. First compute the alerts coming from faulty components.
2. Then compute the alerts coming from non-faulty components.
3. Add them to get the total evidence.

Solution — The evidence is 0.14

About 14% of all components would trigger an alert under this model.

$$\begin{aligned} P(A) &= (0.90)(0.05) + (0.10)(0.95) \\ &= 0.045 + 0.095 = 0.140 \end{aligned}$$

Interpretation

14% of all components will generate an alert.

Why it matters

The denominator normalizes across all explanations for the evidence.

Now update the belief

Once the evidence term is known, the posterior is a direct Bayes update.

Posterior to compute

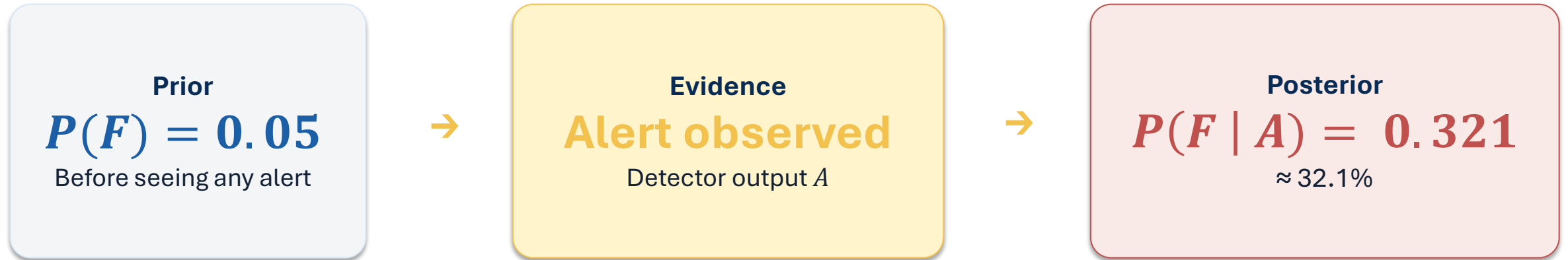
$$P(F | A) = ?$$

$$P(F | A) = \frac{P(A | F) P(F)}{P(A)}$$

$$P(F | A) = \frac{(0.90)(0.05)}{0.14} = \frac{0.045}{0.14}$$

Posterior belief after the alert

The alert matters a lot, but it does not make the component certainly faulty.



Compare:
Belief increased from 5% to about 32%,
so the alert is informative without making the event certain.

Natural-frequency view: imagine 1000 components

Many students find Bayes easier to interpret with counts instead of probabilities.

Out of 1000 components

50 faulty

950 not faulty

Apply detector behavior

45 true alerts

95 false alerts

The same detector behavior applied to a much larger non-faulty population
can still generate many false alerts.

Among all alerts, how many are real faults?

The posterior is the fraction of alerts that came from truly faulty components.

True alerts: 45

False alerts: 95

Total alerts: 140

45 true alerts

95 false alerts

$$P(F | A) = 45 / 140 \approx 0.321$$

This count-based view explains why $P(A | F) = 0.90$ does not imply $P(F | A) = 0.90$.

Why base rates (priors) matter

Posterior belief depends on both the detector and how common the event is to begin with.

Low base rate
Faults are rare

Even a good detector can
produce many false alerts.

Same detector
**Likelihoods
unchanged**

$$P(A | F) = 0.90 \text{ and} \\ P(A | \neg F) = 0.10$$

Posterior
 $\approx 32\%$
after alert

A small false-positive rate applied to a huge normal population can dominate the alert stream.

You Try: what if the base rate changes?

Keep the detector fixed, but make faulty components more common.

Original prior

$$P(F) = 0.05$$

Posterior after alert ≈ 0.321

New prior

$$P(F) = 0.30$$

Predict first: will $P(F | A)$ increase, decrease, or stay the same?

**Think conceptually before calculating:
if faults are more common, an alert should be more credible evidence of a real fault.**

Same detector, different prior

Higher base rate makes the same alert much more convincing.

Prior fault rate $P(F)$	Evidence term $P(A)$	Posterior after alert $P(F A)$
0.05	0.14	0.321 $\approx$ 32%
0.30	0.34	0.794 $\approx$ 79%

Base rates are not optional context. They are part of the inference.

Bayes update checklist

A reliable workflow helps students avoid mixing up the quantities.

- 1 Identify the hypothesis you want to infer.
- 2 Identify the observed evidence.
- 3 Write the prior probability.
- 4 Write the likelihood(s) for the evidence.
- 5 Compute the total evidence using all pathways.
- 6 Apply Bayes' Rule to get the posterior.
- 7 Interpret the posterior in words.

You Try: full Bayes problem

Now complete an update in a new AI scenario.

Fraud detector example

F = transaction is fraudulent

A = detector raises an alert

Given

$$P(F) = 0.08$$

$$P(A | F) = 0.85$$

Find $P(F | A)$.

Show the evidence calculation, the Bayes update, and one sentence interpreting the result.

Solution — Fraud detector update

Structure the arithmetic so each probability plays the correct role.

Evidence:

$$P(A) = (0.85)(0.08) + (0.15)(0.92) = 0.206$$

Posterior:

$$P(F | A) = (0.85 \times 0.08) / 0.206 \approx 0.330$$

Interpretation

If the detector raises an alert, the probability of actual fraud is about 33%.

Common mistake #1

Do not reverse a conditional probability without doing a Bayes update.

Incorrect shortcut

$$P(A | F) = P(F | A)$$

This treats “probability of evidence given hypothesis” as if it were “probability of hypothesis given evidence.”

Correct idea

Use prior + likelihood + evidence

Posterior requires normalization across all explanations for the evidence.

Reversing the condition changes the question. Bayes’ Rule is the tool that connects the two.

Common mistake #2

Detector accuracy is not the same thing as posterior confidence.

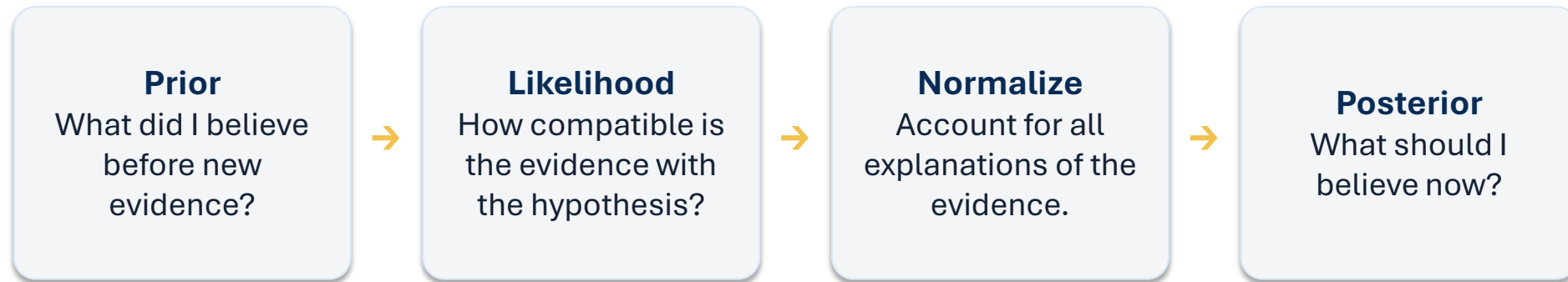
What students often say
**“The detector is 90% accurate,
so the alert means 90%
chance of fault.”**

What Bayes says
Credibility of the alert depends on both
detector behavior and the base rate of the
event.

Always ask:
How rare is the event?
How selective is the evidence?
What other pathways can produce the same observation?

Bayes as belief revision

This same logic will reappear throughout Block 3.



Compact memory aid: $\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$, then normalize.

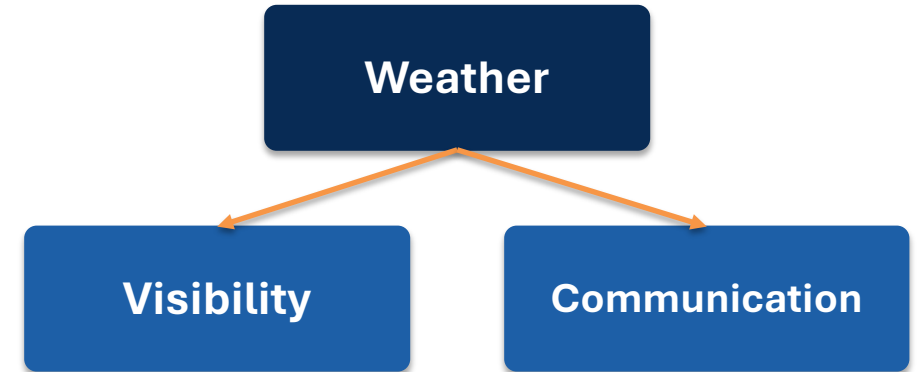
Looking ahead: Bayesian networks

Next lesson: uncertain variables influence one another in a connected graph.

Single-variable update today

Hypothesis → evidence → posterior

One uncertain variable and one observed piece of evidence.



Exit ticket:

(1) What is the prior?

(2) Why can a good detector still produce a modest posterior?

(3) How does a larger base rate affect the posterior?